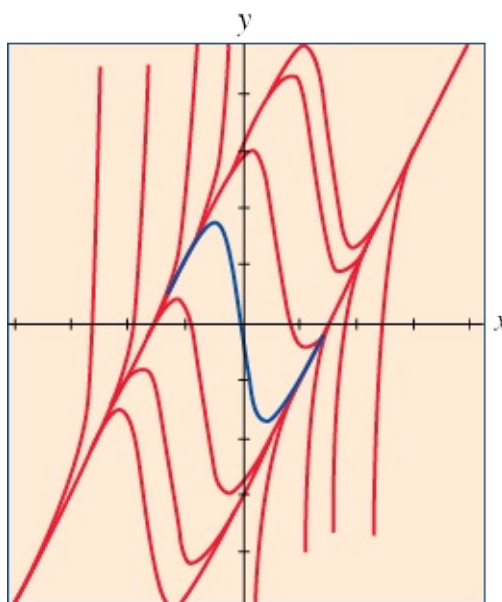


Finally, applying the initial condition  $y(0) = 0$  to the last equation in (6) gives  $c = -1$ . With the aid of a graphing utility we have shown in **FIGURE 2.5.1** the graph of the particular solution

$$y = 2x + \frac{3(1 - e^{6x})}{1 + e^{6x}}$$

in blue along with the graphs of some other members of the family solutions (6). ≡



**FIGURE 2.5.1** Some solutions of the DE in Example 3

## 2.5 Exercises

Answers to selected odd-numbered problems begin on page ANS-3.

*Each DE in Problems 1–14 is homogeneous.*

In Problems 1–10, solve the given differential equation by using an appropriate substitution.

1.  $(x - y) dx + x dy = 0$
2.  $(x + y) dx + x dy = 0$
3.  $x dx + (y - 2x) dy = 0$
4.  $y dx = 2(x + y) dy$
5.  $(y^2 + yx) dx - x^2 dy = 0$

6.  $(y^2 + yx) dx + x^2 dy = 0$
7.  $\frac{dy}{dx} = \frac{y - x}{y + x}$
8.  $\frac{dy}{dx} = \frac{x + 3y}{3x + y}$
9.  $-y dx + (x + \sqrt{xy}) dy = 0$
10.  $x \frac{dy}{dx} = y + \sqrt{x^2 - y^2}, x > 0$

In Problems 11–14, solve the given initial-value problem.

11.  $xy^2 \frac{dy}{dx} = y^3 - x^3, y(1) = 2$
12.  $(x^2 + 2y^2) \frac{dx}{dy} = xy, y(-1) = 1$
13.  $(x + ye^{y/x}) dx - xe^{y/x} dy = 0, y(1) = 0$
14.  $y dx + x(\ln x - \ln y - 1) dy = 0, y(1) = e$

Each DE in Problems 15–22 is a Bernoulli equation.

In Problems 15–20, solve the given differential equation by using an appropriate substitution.

15.  $x \frac{dy}{dx} + y = \frac{1}{y^2}$
16.  $\frac{dy}{dx} - y = e^x y^2$
17.  $\frac{dy}{dx} = y(xy^3 - 1)$
18.  $x \frac{dy}{dx} - (1 + x)y = xy^2$
19.  $t^2 \frac{dy}{dt} + y^2 = ty$
20.  $3(1 + t^2) \frac{dy}{dt} = 2ty(y^3 - 1)$

In Problems 21 and 22, solve the given initial-value problem.

21.  $x^2 \frac{dy}{dx} - 2xy = 3y^4, y(1) = \frac{1}{2}$
22.  $y^{1/2} \frac{dy}{dx} + y^{3/2} = 1, y(0) = 4$

Each DE in Problems 23–30 is of the form given in (5).

In Problems 23–28, solve the given differential equation by using an appropriate substitution.

23.  $\frac{dy}{dx} = (x + y + 1)^2$

24.  $\frac{dy}{dx} = \frac{1 - x - y}{x + y}$

25.  $\frac{dy}{dx} = \tan^2(x + y)$

26.  $\frac{dy}{dx} = \sin(x + y)$

27.  $\frac{dy}{dx} = 2 + \sqrt{y - 2x + 3}$

28.  $\frac{dy}{dx} = 1 + e^{y-x+5}$

In Problems 29 and 30, solve the given initial-value problem.

29.  $\frac{dy}{dx} = \cos(x + y), \quad y(0) = \pi/4$

30.  $\frac{dy}{dx} = \frac{3x + 2y}{3x + 2y + 2}, \quad y(-1) = -1$

## Discussion Problems

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Some second-order differential equations can be reduced to a first-order equation by means of a substitution. In Problems 31 and 32, use the substitution  $u = y'$  to reduce the given differential equation to a first-order equation, and then solve that equation by a method discussed in this section. Then use  $u = y'$  again to find the solution  $y$  of the given equation.

31.  $y'' + (y')^2 = 0$

32.  $y'' - 2y' = 4x$

33. (a) Show that it is always possible to express any homogeneous differential equation  $M(x, y) dx + N(x, y) dy = 0$  in the form

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right).$$

(b) Show that the substitution  $u = y/x$  in the last differential equation in part (a) yields a separable equation.

34. Put the homogeneous differential equation

$$(5x^2 + 2y^2) dx - xy dy = 0$$

in the form  $dy/dx = F(y/x)$  given in part (a) of Problem 33.

Then solve the equation using the substitution  $u = y/x$ .

35. (a) Determine two singular solutions of the DE in Problem 10.

(b) If the initial condition  $y(5) = 0$  is as prescribed in Problem 10, then what is the largest interval  $I$  over which the solution is defined? Use a graphing utility to plot the solution curve for the IVP.

36. In Example 3, the solution  $y(x)$  becomes unbounded as  $x \rightarrow \pm \infty$ . Nevertheless  $y(x)$  is asymptotic to a curve as  $x \rightarrow -\infty$  and to a different curve as  $x \rightarrow \infty$ . Find the equations of these curves.

37. The differential equation

$$\frac{dy}{dx} = P(x) + Q(x)y + R(x)y^2$$

is called a **Riccati equation** after the Venetian jurist and mathematician Jacopo Francesco Riccati (1676–1754) who was able to solve several DEs of this type.

(a) A Riccati equation can be solved by a succession of two substitutions *provided* we know a particular solution  $y_1$  of the equation. Show that the substitution  $y = y_1 + u$  reduces Riccati's equation to a Bernoulli equation (4) with  $n = 2$ . The Bernoulli equation can then be reduced to a linear equation by the substitution  $w = u^{-1}$ .

(b) Find a one-parameter family of solutions for the differential equation

$$\frac{dy}{dx} = -\frac{4}{x^2} - \frac{1}{x}y + y^2,$$

where  $y_1 = 2/x$  is a known solution of the equation.

38. Devise an appropriate substitution to solve

$$xy' = y \ln(xy).$$

## Mathematical Models

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39. **Population Growth** In the study of population dynamics one of the most famous models for a growing but bounded population is the **logistic equation**

$$\frac{dP}{dt} = P(a - bP),$$

where  $a$  and  $b$  are positive constants. Although we will come back to this equation and solve it by an alternative method in Section 2.8, solve the DE this first time using the fact that it is a Bernoulli equation.

40. **Falling Chain** In Problem 47 in Exercises 2.4 we saw that a mathematical model for the velocity  $v$  of a chain slipping off the edge of a high horizontal platform is

$$xv \frac{dv}{dx} + v^2 = 32x.$$

In that problem you were asked to solve the DE by converting it into an exact equation using an integrating factor. This time solve the DE using the fact that it is a Bernoulli equation.